

MCT344 / MCT342 – Industrial Robotics - Project Description

Mechatronics and Robotics Engineering Program

Course Project: Modeling, Simulation, and Analysis of an Industrial Robotic Manipulator

1. Project Overview

This project aims to develop a comprehensive analytical and simulation-based study of an industrial robotic manipulator (5-DOF, 6-DOF, or 7DoF). Students will model, analyze, simulate, and evaluate the robotic system using theoretical foundations from the course and advanced computational simulation tools.

The project integrates mathematical modeling, numerical computation, simulation, and control design to provide a complete understanding of industrial robotic systems.

2. Robot Selection

Each team must select **one** of the following robotic systems:

- **ABB IRB 120** (The robot that you observed in the laboratory)
- **Franka Emika Panda Robot** (The robot that you observed in the laboratory)
- The robot developed in the *Design of Mechatronic Systems* course by your team

In case of ABB IRB 120 and Franka Emika Panda Robot, all robot parameters, CAD, and specifications must be obtained from official datasheets or reliable technical resources.

Each team must clearly justify their selection and document the robot specifications.

3. Project Objectives

The selected robotic system must be modeled and analyzed through the following components:

A) Kinematic Analysis

- Denavit–Hartenberg (DH) parameter assignment
- Forward Kinematics (Homogeneous transformation matrices)
- Inverse Kinematics

- Verification of the analytical kinematic analysis through comparison with Robotics System Toolbox™ results.

B) Workspace Analysis

- Reachable workspace
- Dexterous workspace
- Visualization of the workspace

C) Jacobian Analysis

- Analytical derivation of the Jacobian matrix
- Singularities analysis
- Velocity mapping (joint space $\leftrightarrow$ task space)

D) Dynamic Modeling

- Derivation of robot dynamics using:
 - Euler–Lagrange formulation or
 - Newton–Euler method
- Inertia matrix
- Coriolis and centrifugal terms
- Gravity vector
- Validation using simulation tools

E) Trajectory Planning

- Joint-space trajectory planning
- Cartesian trajectory planning
- Polynomial trajectories (cubic / quintic)
- Velocity and acceleration constraints
- Smooth motion generation

F) Control Design

- Joint-space control (e.g., PD, PID, Computed Torque Control)

- Performance evaluation (tracking error, stability)
- Simulation-based validation

G) Task Simulation

Assume a realistic industrial application. You may select one of the tasks listed below or propose an alternative task, such as:

- Pick-and-place
- Assembly
- Welding trajectory
- Surface tracking trajectory
- Material handling

You must:

- Plan trajectories
- Simulate control
- Evaluate performance

4. Modeling and Simulation Requirements

The project must be implemented using three different approaches, followed by a comprehensive comparison of the results obtained from each method:

Method 1: Analytical Modeling (From Scratch)

- Develop full mathematical models manually using equations derived in the course
- Solve and simulate using MATLAB scripts (m-files)
- No use of built-in Robotics Toolbox functions for this part
- Ensure that all components outlined in the project objectives are thoroughly completed and analyzed

Method 2: MATLAB Robotics System Toolbox™ (m-files)

- Build the robot model using:
 - rigidBodyTree

- inverseKinematics
 - geometricJacobian
 - inverseDynamics
 - trajectory generation functions
- Ensure that all components outlined in the project objectives are thoroughly completed and analyzed
- Perform kinematic (Forward/Inverse), dynamic (Forward/Inverse), and trajectory analysis
- Compare results with analytical modeling

Method 3: Simulink + Simscape + Robotics System Toolbox

- Model the robot in:
 - Simulink
 - Simscape Multibody
 - Robotics System Toolbox blocks
- Implement:
 - Ensure that all components outlined in the project objectives are thoroughly completed and analyzed
 - Motion planning
 - Control system
 - Dynamic simulation
- Visualize robot motion

Comparison Requirement

You must compare:

- Analytical results
- MATLAB Toolbox results obtained using mfiles
- Simulink/Simscape results

Discuss:

- Accuracy
- Numerical differences
- Modeling complexity

5. Experimental Bonus (4 Extra Marks)

Teams that select the robot developed in the *Design of Mechatronic Systems* course **may receive up to 4 bonus marks** if they:

- Implement inverse kinematics experimentally
- Apply robotic control techniques on the real robot
- Demonstrate successful trajectory planning and tracking
- Show recorded experimental results
- Compare experimental results with simulation

The robot experiments must function correctly and demonstrate proper control performance in compare to simulation.

6. Deliverables

A) Technical Report (Submitted as PDF file)

A detailed professional technical report must include:

1. Robot description and specifications
2. DH parameters and modeling assumptions
3. Full analytical derivations
4. Mathematical equations and explanations
5. Simulation methodology
6. Results and plots
7. Task simulation results
8. Control performance analysis

9. Comparative study between methods

10. Discussion and conclusions

The report must be well-organized, clearly formatted, and technically rigorous.

B) MATLAB and Simulation Files (submitted as soft copy)

Submit:

- All MATLAB scripts (.m files)
- Simulink models (.slx files)
- Simscape files
- Organized folder structure
- Clear naming convention
- README file as PDF or word document explaining how to run the project

C) Project Video (5–10 Minutes)

Each team must submit a narrated video including:

- Introduction of the robot
- Explanation of modeling steps
- Demonstration of simulations
- Trajectory planning results
- Control results
- Comparison between methods
- Commentary on observations

The video must include simulation visuals and explanation of results.

d) Team responsibility sheet

7. Submission Deadline

The complete project must be submitted in: **Week 14**

8. Project Team Formation and Academic Integrity

- Each project team must consist of 3 to 4 students.
- A maximum of 4 students per team is allowed. Teams exceeding this limit will not be accepted.
- Academic integrity is strictly enforced. Any form of copying between teams will result in a zero grade for the project for all involved teams.
- Plagiarism is strictly prohibited. All work must be original and developed by the team members.
- Each team must submit a Responsibility Statement clearly indicating:
 - The role of each team member
 - The specific tasks completed by each member
 - Individual contributions to modeling, simulation, analysis, coding, and reporting

Failure to submit the responsibility sheet may affect the project evaluation.